

EXPERIENCE

Hardware Specialist Apple Inc. Corvallis, OR 2016-18

* College Program

Logistics Specialist UPS Inc. Corvallis, OR 2016-18

* College program

College Networking & Computer Analyst Oregon State University: College Of Science Corvallis, OR 2016

* Liaison between IT Department and the College Of Science; assisted PHD/MSc students and research cadre with issues.

* Assisted interfacing with LabView, internal tools, scientific equipment, lasers, electron microscopes, particle analyzers, and rf tools.

Fire Support Specialist Paratrooper 13F1PJ7 US Army Ft. Bragg, NC 2011

* Communications via satellite and terrestrial RF using Encrypted Trunking Data/Voice as a support element for maneuver commanders; assisted with fire support from remote assets after presenting fire plans while following fire support guidelines.

Networking Engineer Olypen Inc. Sequim, WA 2002

* Automation of System Administration tasks and Network security analysis and mapping using unix.

* Penetration testing and security improvements * Perl/Bash/Unix Automation tool development

EDUCATION

Oregon State University Electrical/Computer Engineering Senior

* ABET accredited engineering program

* Dual focus in Electromagnetism and Networking Junior level

SKILLS

Computer Engineering

* Data structures (C/C++)

* CAD Eagle/KiCad

* PCB Manufacturing

* Interfacing I2C/Serial/BT

* Object Oriented Programming

* Automation using off-shelf components

* Virtualization & Emulation

* Algorithms

* Boolean logic minimization

* Networking Topologies/Congestion

* Digital Communications

* Python Panda/Matplotlib/File/Fork

* Matlab/R/Python Data

* Cross-Compiling Toolchains

* Object Oriented Design

* Architecture (ARM/MIPS/RISC)

* ASM RISC

* Microcontrollers

* Unix Development

* Network Socket Programming (IP4/IP6)

* Unix/Linux/BSD/Mach/Solaris

Electrical Engineering

* Transmission lines

* Analog filters

* Electrostatics/magnetostatics

* Discrete devices

* Wave propagation

* Electromagnetic compatibility

PROJECTS

Aircraft Location Verification System

* Pointing multiple directional antennas in the direction of planes flying through the atmosphere for verification of ADSB location broadcasts; verified demodulated data with actual aircraft location via stepper/servos and modulation techniques.

* Interfacing Location data with remote devices through network using vector geospatial data rendered via geodesic to curvature calculation in the Python programming language; broadcasted data in an interchangeable format synchronously over TCP/IP.

* Designing and implementing a socket based tcp based program for pulling data over a network and updating to a disk simultaneously.

* Automated kiosk operation using web based services locally on Chromium based browser using browser extensions and build options.

* Demodulating ADSB Type A/B signals then providing overlay data for use on cached maps locally in Python, javascript and SQL.

Network Security and Mapping Automation

* Using bash scripting and perl log rotation tools that using output of networking mapping unix tools displaying the output via vector graphics; printing the outputting vector based maps. Automated data representation/presentation for cost savings.

* Implemented intrusion detection system and penetration tools using snort, nmap scripts, and bandwidth throttling using ipfw.

* Configuration file generation and automated uploading to Juniper, Cisco and other BSD devices for infrastructure scaling.

Home Automation System Implementation

* Automated Home surveillance system using Apache/SSL on a Linux system using custom embedded firmware on off the shelf wifi cameras for cost reduction. Cross-compiling toolchain use for developing custom firmware.

* Implemented different software video encoding techniques as a workaround for closed source hardware encoding chips.

Joule Stealing Flashlight PCB replacement

* PCB utilized to replace a 3V LED flashlight with a lower volt 1.5V source by modulating the light output while using an induction circuit's greater amperage to buck boost voltage; installing PCB into a store counter flashlight after cost analysis.

